Kinetic energy and net work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net work to a change in kinetic energy.

SUCCESS CRITERIA - I can...

1. Calculate kinetic energy with speed squared.
2. Use net work rather than one force's work.
3. Connect the sign of work to a speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Keep mass fixed and multiply speed by 3. By what factor does kinetic energy change? Explain from the equation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $K = \frac{1}{2}mv^2$; $W_{net} = K_f - K_i$.
i = initial; f = final. Positive net work increases kinetic energy.

Worked reasoning: A 2 kg cart at 2 m/s has K=4 J. Adding 5 J of net work gives final K=9 J, so final speed is 3 m/s.

Your turn: Draw before-and-after energy bars and label the transfer between them.

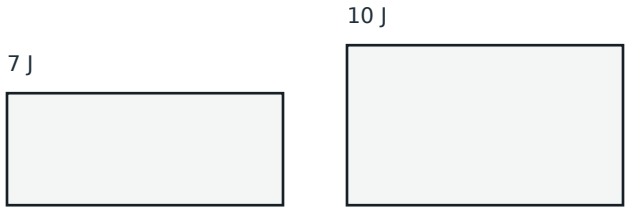
Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

A lab cart's kinetic energy changes from 7 J to 10 J, as shown. Find the net work. Explain how its speed changes if mass stays constant.



Marzano check - circle: ① ② ③

04 Apply the physics

A 2 kg cart moves at 3 m/s. A pull and all other forces together do +7 J of net work. Find its initial and final kinetic energy. Complete the energy account.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student adds the work of a pull but ignores friction. Why might their predicted speed be too high?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two 4 kg carts latch together after colliding. Is this perfectly inelastic? State what remains conserved if external impulse is negligible.
Use before and after snapshots to organize the momentum account.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

K changes from 20 J to 8 J. Find net work and state whether speed rises or falls. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Keep the cart's mass fixed. Compare the energy needed to increase speed from 1 to 2 m/s with that from 2 to 3 m/s. Use m=2 kg. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3